

Armaan Kanchan

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SUMMARY

Entry-level Software Tester / QA Engineer with hands-on experience in system validation, test automation, networked systems, and embedded firmware testing. Strong background in defect tracking, test case creation, log analysis, and cross-team debugging in fast-paced environments.

EDUCATION

Purdue University West Lafayette, IN
B.S. in Computer Engineering May 2026
Relevant: Computer Communication Networks, Operating Systems, Linux, Embedded Systems, Digital Design

EXPERIENCE

SanDisk Corporation Milpitas, CA
Software / Systems Test Intern May 2025 – Aug 2025

- Designed and executed **system-level test scripts (Linux shell)** to validate flash DUT connectivity, kernel/IP behavior, SPI, USB, and PCIe link width/speed (Gen 3/4/5).
- Detected, reproduced, and documented software and interface defects using logs, diagnostics, and repeatable test procedures.
- Developed a **Qt (C++) test GUI** integrating multiple validation workflows, improving test consistency and reducing manual execution errors.
- Built an **Ansible-based deployment and configuration framework** for prod/QA/staging environments, enabling consistent test environments and faster release validation.

Western Digital Corporation San Jose, CA
Test Platform Development Intern May 2024 – Aug 2024

- Engineered an **automated QA test infrastructure** for an internal diagnostic platform, integrated with **Jenkins CI/CD** for continuous testing.
- Implemented **Playwright-based UI and system test suites** to validate workflows across builds and catch regressions early.
- Reduced manual QA effort by automating end-to-end test scenarios; improved release reliability and turnaround time.
- Collaborated with developers to analyze test failures, isolate root causes, and verify fixes through regression testing.

PROJECTS

GoldenForm: Wearable IMU System (ESP32, BLE, microSD) Aug 2025 – Present
Purdue University

- Designed and executed **functional and stress tests** for BLE connectivity, sensor sampling accuracy, and data integrity under disconnects and power loss.
- Validated embedded firmware behavior using logs and repeatable test scenarios; identified edge cases related to timing, buffering, and storage failures.
- Tested end-to-end data flow: IMU acquisition → on-device logging/streaming → host-side analysis.

Computer Communication Networks Labs 2025
Purdue University

- Implemented and tested networked applications using structured test cases to validate client/server behavior and protocol correctness.
- Analyzed logs and packet-level behavior to diagnose latency, ordering, and connection issues.

AHB-Lite USB 1.0 Verification (SystemVerilog) Jan 2025 – May 2025
Purdue University

- Built SystemVerilog testbenches and **QuestaSim** flows to validate USB bulk transfers, error handling, and AHB-Lite protocol compliance.
- Verified data integrity, timing, and bus responses under normal and fault conditions.

SoCET - Embedded Systems QA & Validation 2024 – 2025
Purdue University

- Performed **QA testing and validation** for SPI and **QSPI**-based peripheral communication on microcontroller platforms.
- Designed and executed **test cases** to verify correct clock polarity/phase (CPOL/CPHA), data ordering, chip-select timing, and multi-slave behavior.
- Validated **QSPI flash memory transactions** including read/write/erase operations, burst transfers, and error recovery under timing and signal variations.
- Used logic analyzers and firmware logs to **detect, reproduce, and debug communication defects** such as misaligned frames, timing violations, and data corruption.
- Collaborated with firmware developers to analyze test failures, suggest fixes, and perform **regression testing** after updates.

SKILLS

Testing/QA: Test case design, regression testing, defect tracking, log analysis, test documentation

Networking: TCP/IP fundamentals, client/server testing, BLE concepts (GATT, services)

Languages: C/C++, Python, Bash, SystemVerilog

Tools: Linux, Git, Jenkins, Playwright, Qt Creator, Ansible, QuestaSim

Embedded: ESP32, SPI, I2C, USART, USB, GPIO